

SUMMATIVE ASSESSMENT - II

MATHEMATICS

General Instruction:

- i) All Questions are Compulsory
- ii) The question paper consists of 31 questions divided into four sections - A, B, C and D.
- iii) Section A contains 4 questions of 1 mark each. Section B contains 6 questions of 2 marks each, Section C contains 10 questions of 3 marks each and Section D contains 11 questions of 4 marks each.
- iv) Use of calculators is not permitted

SECTION - A

Question numbers 1 to 4 carry 1 mark each.

- 1. A ladder, leaning against a wall, makes an angle of 60° with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder.
- 2. For what value of k will $k + 9$, $2k - 1$ and $2k + 7$ are the consecutive terms of an A.P.?
- 3. In Fig.1, PQ is a tangent at a point C to a circle with centre O. If AB is a diameter and $\angle CAB = 30^\circ$, find $\angle PCA$.

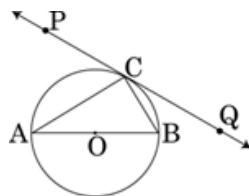


Figure 1: Fig 1

- 4. A card is drawn at random from a well shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen.

SECTION - B

Question numbers 5 to 10 carry 2 marks each.

5. In Fig.2, a quadrilateral ABCD is drawn to circumscribe a circle with centre O in such a way that the sides AB, BC, CD and DA touch the circle at points P, Q, R and S respectively. Prove that

$$AB + CD = BC + DA$$

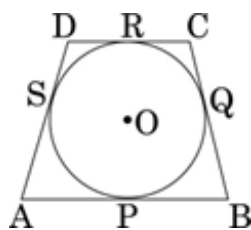


Figure 2: Fig 2

6. The 4th term of an A.P. is zero. Prove that the 25th term of the A.P. is three times its 11th term.
7. In Fig.3, from an external point P, two tangents PT and PS are drawn to a circle with centre O and radius r . If $OP = 2r$, show that

$$\angle OTS = \angle OST = 30^\circ$$

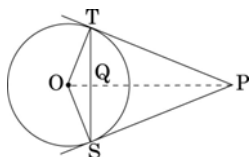


Figure 3: Fig 3

8. Prove that the points $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right angled isosceles triangle.
9. Let P and Q be the points of trisection of the line segment joining the points $A(2, -2)$ and $B(-7, 4)$ such that P is nearer to A. Find the coordinates of P and Q.
10. Solve for x :

$$\sqrt{2x + 9} + x = 13$$

SECTION - C

Question numbers 11 to 20 carry 3 marks each.

11. If the point $P(x, y)$ is equidistant from the points $A(a+b, b-a)$ and $B(a-b, a+b)$, prove that

$$bx = ay$$

12. A conical vessel with base radius 5 cm and height 24 cm is full of water. This water is emptied into a cylindrical vessel of base radius 10 cm. Find the height to which the water will rise in the cylindrical vessel.

$$\left(\text{Use } \pi = \frac{22}{7} \right)$$

13. In Fig.4, O is the centre of a circle such that diameter $AB = 13$ cm and $AC = 12$ cm. BC is joined. Find the area of the shaded region.

$$(\text{Take } \pi = 3.14)$$

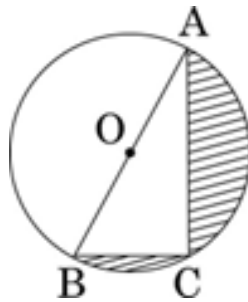


Figure 4: Fig 4

14. A sphere of diameter 12 cm is dropped into a right circular cylindrical vessel partly filled with water. If the sphere is completely submerged in water, the water level rises by

$$3\frac{5}{9} \text{ cm}$$

Find the diameter of the cylindrical vessel.

15. In Fig.5, a tent is in the shape of a cylinder surmounted by a conical top of the same diameter. If the height and diameter of the cylindrical part are 2.1 m and 3 m respectively, and the slant height of the conical part is 2.8 m, find the cost of canvas needed to make the tent if the canvas is available at the rate of Rs.500.

$$\left(\text{Use } \pi = \frac{22}{7} \right)$$

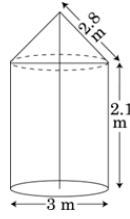


Figure 5: Fig 5

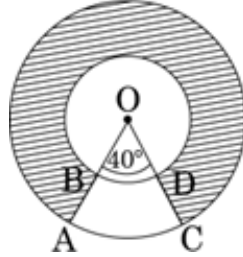


Figure 6: Fig 6

16. In Fig.6, find the area of the shaded region enclosed between two concentric circles of radii 7 cm and 14 cm where $\angle AOC = 40^\circ$.

$$\left(\text{Use } \pi = \frac{22}{7} \right)$$

17. A man standing on the deck of a ship, which is 10 m above water level, observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of hill as 30° . Find the distance of the hill from the ship and the height of the hill.
18. The digits of a positive number of three digits are in A.P. and their sum is 15. The number obtained by reversing the digits is 594 less than the original number. Find the number.
19. If the roots of the quadratic equation

$$(a - b)x^2 + (b - c)x + (c - a) = 0$$

are equal, prove that

$$2a = b + c$$

20. From a pack of 52 playing cards, Jacks, Queens and Kings of red colour are removed. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is:
- (a) a black King
 - (b) a card of red colour
 - (c) a card of black colour